

Adjoint Sensitivities and Gradient-Based Manning Roughness Calibration in RDycore

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Abstract

We present a discrete-adjoint capability for RDycore, the exascale river dynamical core of the Energy Exascale Earth System Model (E3SM), and use it for gradient-based calibration of spatially distributed Manning roughness. The capability has three parts. First, an exact assembled Jacobian of the finite-volume shallow-water right-hand side: closed-form Roe-flux blocks obtained by hand-written forward-mode differentiation of the flux computation, entropy-fix branches included, together with source-term and boundary-condition blocks. Second, adjoint sensitivities of gauge-misfit functionals with respect to both the initial state and the Manning field via PETSc’s `TSAdjoint`; one gradient costs roughly one extra forward solve, independent of the parameter count. Third, bound-constrained calibration with TAO. Every derivative is validated against finite differences of the full nonlinear solve at tight, pre-registered tolerances (Jacobian blocks to 10^{-8} , gradients to 10^{-5}). Twin experiments recover a two-zone roughness field to 4×10^{-7} relative error under noise-free observations — a statement about gradient correctness, not about calibratability, and we quantify the difference. The same assembled Jacobian enables fully implicit backward-Euler stepping, and we show implicit stepping is required in practice: explicit Manning drag is unstable at the operational time step of a Hurricane Harvey test case, in quantitative agreement with a friction-stiffness stability analysis. The implicit integrator’s adjoint passes the same finite-difference gates. On a Hurricane Harvey twin configuration (2,746 cells, implicit stepping) the calibrated Manning map recovers a two-zone truth field to 19.4% with proper Tikhonov regularization. The gauge-sparse setting exhibits — and lets us dissect — the classic semiconvergence of underdetermined friction estimation: a twin using the actual 17-gauge USGS network geometry fits its 238 observations essentially perfectly while recovering the field to only 45%, quantifying why real-data calibration must shrink the parameter space to match the network — and, re-asking identifiability at basin scale, that the observable matters as much as the parameterization: peak water-surface elevations at 108 surveyed high-water-mark locations identify the land-cover roughness classes covering 83% of a 2.93M-cell Hurricane Harvey domain, where 13 stream gauges identify almost none. Within E3SM, adjoint-based parameter optimization is established in the land-ice component MALI, whose velocity solver inverts for basal friction by automatic differentiation of the *steady* momentum balance; RDycore’s capability is, to our knowledge, the first *transient* discrete adjoint

in an E3SM component — differentiated through the time integrator itself. It is obtained by differentiating the existing solver in place: the production discretization, its data structures and its GPU backends are the ones being differentiated, not a reimplementaion of the model in a differentiable-programming framework.

1 Introduction

Manning roughness is the dominant tunable parameter of depth-averaged flood models, and its calibration is traditionally manual or ensemble based. Adjoint methods deliver the gradient of an observation misfit with respect to *every* cell’s roughness at the cost of about one extra forward solve, making distributed calibration tractable. The approach is well established in shallow-water modeling, from channel networks [1] and unstructured finite-volume twin experiments [2] to GPU-accelerated optimizers [3]. It is deployed in variational-assimilation platforms such as DassFlow [4, 5] and the Thetis coastal model, which calibrated a spatially varying Manning field for the Baltic Sea with an automatically generated adjoint [6], following storm-surge sensitivity studies that addressed the overfitting inherent in distributed friction estimation [7].

A recent generation of *differentiable* flood and hydrology codes obtains gradients by rewriting simplified solvers inside machine-learning frameworks: Inunda (PyTorch, local-inertial, calibrated on a Hurricane Harvey hindcast) [8], Hydrograd (Julia) [9], AegirJAX [10], CaMa-Flood-GPU [11], and differentiable routing [12]. These demonstrate demand for exactly this capability, but none is a production Earth-system component: they trade the validated numerics, unstructured meshes, and coupling infrastructure of codes like RDycore for differentiability.

This paper takes the complementary route: make the production code differentiable. The route has precedent inside E3SM itself: the land-ice component MALI [13] routinely inverts for basal friction with an adjoint obtained by automatic differentiation of its Albany velocity solver [14] — but that is the adjoint of a *steady* nonlinear solve, used for initialization; no E3SM component has, to our knowledge, a discrete adjoint of its transient time stepper. RDycore is a finite-volume shallow-water dynamical core (Roe fluxes, well balancing, wetting and drying) validated at up to 471M cells. We add the missing derivative infrastructure — an exact assembled Jacobian and TSAdjoint-based sensitivities [15] — entirely behind a configuration flag, with no impact on existing runs and no new dependencies.

Contributions: (i) an exact, FD-verified Jacobian of the SWE right-hand side, including Roe dissipation, entropy-fix branches, wet/dry branches, and reflecting/Dirichlet/critical-outflow boundary conditions; (ii) adjoint gradients dJ/du_0 and dJ/dn with respect to water-height observations at gauge locations at multiple observation times, FD-validated to 10^{-5} or better; (iii) TAO-based recovery of per-region, per-cell and land-cover-class roughness fields, with a quantified study of when the observing network makes them identifiable at all; (iv) implicit (backward Euler) stepping and its adjoint, enabled by the same Jacobian, with an empirical demonstration that explicit friction fails at operational time steps on a Hurricane Harvey configuration; and (v) a catalog of the non-differentiable structures encountered — useful, we believe, to anyone retrofitting adjoints onto an operational flood code.

2 Forward model

RDycore solves the 2D shallow water equations in conservative variables $u = (h, q_x, q_y)$, $\mathbf{q} = h\mathbf{v}$:

$$\partial_t h + \nabla \cdot \mathbf{q} = Q_r(x, t), \quad (1)$$

$$\partial_t \mathbf{q} + \nabla \cdot \left(\frac{\mathbf{q} \otimes \mathbf{q}}{h} + \frac{gh^2}{2} \mathbb{I} \right) = -gh \nabla z_b - gn^2 h^{-7/3} \mathbf{q} \|\mathbf{q}\|, \quad (2)$$

with cell-centered finite volumes on an unstructured mesh:

$$\frac{du_i}{dt} = -\frac{1}{|\Omega_i|} \sum_{e \in \partial\Omega_i} \mathcal{F}(u_i, u_j, \mathbf{n}_e) L_e + S(u_i; n_i, t) \equiv f_i(u, n), \quad (3)$$

where $\mathcal{F}$ is the Roe flux with a critical-flow (entropy) fix, primitives are reconstructed with a regularized division and a dry cutoff $h_{\min}$, and S contains bed slope, Manning drag, and external forcing.

A differentiable right-hand side. A prerequisite that deserves emphasis: RDycore’s existing friction treatments (a semi-implicit splitting and the implicit scheme of Xia and Liang) embed the time step — and, in the semi-implicit case, the flux divergence — *inside* the RHS callback. The resulting map is consistent only with forward Euler and admits no well-defined $\partial f / \partial u$. We therefore added a plain explicit source treatment (`source.method: explicit`), used on every adjoint path; Section 6 shows how the resulting stiffness is then handled properly, by the integrator rather than inside f .

3 Exact Jacobian

The RHS Jacobian is block sparse with 3×3 blocks on the cell-adjacency graph: each interior edge (i, j) contributes $\mp(L_e/|\Omega_{i,j}|) \partial \mathcal{F} / \partial u_{L,R}$ to the two cell rows, local sources contribute diagonal blocks, and boundary edges contribute diagonal blocks through the ghost-state map of the boundary condition.

Flux blocks by forward-mode differentiation. Rather than deriving the dissipation-matrix derivatives symbolically, we differentiate the flux *computation*: a hand-written forward-mode (tangent) version of the Roe flux routine propagates directionals through every intermediate — Roe averages, eigenvectors, eigenvalues including both branches of the critical-flow fix, wave strengths, and physical fluxes — taking the code’s own branch at every $|\cdot|$ and switch (a consistent one-sided derivative). Seeding unit directions through the primitive-reconstruction chain rule yields the two conservative-space blocks in six tangent evaluations per edge. The approach mirrors the flux code line by line, which makes it reviewable and, in our experience, correct on first assembly against the verification harness below.

Boundary blocks. For a boundary edge, $F(u_{\text{in}}, u_g(u_{\text{in}}))$ contributes $(\partial_L F + \partial_R F \partial u_g / \partial u_{\text{in}})$. The reflecting map is a linear mirror; Dirichlet ghosts are constant; critical outflow has a closed-form ghost map $h_g = (q^2/g)^{1/3}$ with the code’s inflow branch producing an identically zero flux.

Parameter Jacobian. Manning n appears only in the drag term, so $\partial S_{\mathbf{q}} / \partial n = -2gnh^{-7/3} \mathbf{q} \|\mathbf{q}\|$: two nonzeros per cell, supplied to `TSSetRHSJacobianP`.

Check	Configuration	Result	Gate
Edge flux blocks	wet jump 10 m/5 m	7.6×10^{-9}	10^{-6}
Edge flux blocks	transcritical (entropy-fix branch)	2.4×10^{-10}	10^{-5}
Source block	shallow flowing state	$< 10^{-6}$	10^{-6}
Assembled matrix	uniform flow, reflecting BCs	1.6×10^{-8}	10^{-6}
Assembled matrix	dam break, reflecting BCs	1.5×10^{-8}	10^{-6}
Assembled matrix	dam break, Dirichlet + critical outflow	1.9×10^{-8}	10^{-6}
dJ/du_0 (explicit RK)	8 sampled components	1.0×10^{-8}	10^{-5}
dJ/dn (explicit RK)	domain aggregate	2.9×10^{-6}	10^{-5}
dJ/du_0 (implicit BE)	tight inner tolerances	1.4×10^{-8}	10^{-5}
dJ/dn (implicit BE)	tight inner tolerances	5.3×10^{-6}	10^{-5}

Table 1: Relative errors of analytic derivatives against central finite differences of the full nonlinear computation. Every check in this table is an automated test in RDycore’s `ctest` suite, run on every change to the repository.

Verification. Three layers, all against central finite differences with pre-registered tolerances (Table 1): an edge-level harness that differences the flux routine itself on sampled states (with a Richardson self-check of the FD order); full-matrix comparisons of the assembled Jacobian against a matrix-coloring FD Jacobian of the actual RHS on three state/boundary combinations, boundary rows included; and end-to-end gradient checks through the adjoint (next section). A finite-difference coloring Jacobian (`numerics.jacobian: fd`) remains available permanently as the verification baseline.

4 Adjoint sensitivities

For gauge observations y_k at times t_k with observation operator H (one row per gauge, selecting water height) and error variance σ^2 ,

$$J(n) = \frac{1}{2\sigma^2} \sum_k \|Hu(t_k) - y_k\|^2. \quad (4)$$

`TSAdjoint` [15] differentiates the discrete time stepper, so gradients are exact to roundoff for the computed trajectory — the property that makes the FD gates in Table 1 meaningful. Multi-time observations are handled by a segmented backward sweep: the adjoint is advanced between observation times with `TSAdjointSetSteps` and receives a jump $H^T \sigma^{-2} r_k$ at each t_k (Algorithm 1 in Section 5). One sweep yields both dJ/du_0 (state sensitivity) and dJ/dn (per-cell parameter gradient, via the two-nonzero parameter Jacobian).

Two integrator facts constrain the design. PETSc’s plain forward-Euler integrator implements no adjoint, so adjoint-path configurations use the Runge–Kutta family. And for implicit integrators the discrete-adjoint gradient is only as exact as the inner solves: with default SNES/KSP tolerances the Manning gradient checked at 2×10^{-4} ; tightening the tolerances recovered 5×10^{-6} (Table 1). We state the production side of this plainly, because the verification numbers are otherwise easy to over-read: at the loose inner tolerance used for the large implicit runs (Krylov relative tolerance 10^{-2} , chosen because it is $\sim 2.5\times$ cheaper end to end), the gradient is inner-solve-limited at 6×10^{-4} – 4×10^{-3} relative. In every calibration we have run, the loose- and tight-tolerance settings converge indistinguishably (same objective to 5–6 digits), consistent with the tolerance of quasi-Newton methods to bounded gradient error; our policy is to run campaigns loose and repeat one calibration at Krylov tolerance 10^{-3} — where the 10^{-5} gates pass — as a spot check.

5 The calibration algorithm: strong-constraint 4D-Var

The complete calibration loop is summarized in Algorithms 1–3. It is a strong-constraint 4D-Var iteration: the model is enforced exactly (the only controls are the Manning field and, optionally, the initial state). Each objective evaluation performs one checkpointed forward sweep that records the gauge residuals r_k at the observation times, followed by one backward sweep of the discrete adjoint (Algorithm 2). The per-iteration cost is therefore one forward plus one adjoint solve, independent of the number of parameters.

The backward sweep carries two vectors with distinct roles. The *adjoint state* λ holds $\partial J_{\text{mis}}/\partial u(t)$ at the current point of the backward traversal. It is seeded with the terminal observation sensitivity $\sigma^{-2}H^T r_K$, propagated one step at a time by the transposed Jacobian of the forward step map (the backward “model step”), and incremented by a jump $\sigma^{-2}H^T r_k$ each time the sweep crosses an observation time, because the state at that time enters the misfit directly. The *parameter sensitivity* μ starts at zero and accumulates, at every step, the chain-rule contribution $(\partial\Phi_s/\partial n)^T \lambda$ of that step’s Manning dependence. When the sweep reaches $t = 0$, μ is the complete misfit gradient dJ_{mis}/dn and λ is dJ_{mis}/du_0 . Both transposed Jacobians are the analytic operators of Sections 3–4 — the assembled RHS Jacobian for the state propagation, the two-nonzero parameter Jacobian for the μ accumulation — applied within TSAdjoint’s discrete adjoint of the chosen integrator.

The outer update is TAO’s bound-constrained limited-memory quasi-Newton method BLMVM [16] (Algorithm 3). The loop terminates on the standard bound-constrained test: the projected gradient g^P — the gradient with components pointing out of the feasible box zeroed — drops below an absolute tolerance, or an iteration budget is reached. The same loop serves every parameterization in this paper: per-region, per-cell, and land-cover-class fields differ only in the expansion of the parameter vector onto cells (line 3 of Algorithm 1) and the corresponding summation of μ back onto parameters.

Practical execution. One objective evaluation costs one checkpointed forward sweep plus one adjoint sweep, independent of the number of parameters; each trial step of the BLMVM line search costs one such evaluation. A basin-scale window needs more iterations than a single batch allocation provides, so a run writes its parameter vector on exit and the next reads it back as its starting iterate, matched by land-cover class rather than by position. The resume is exact in the parameters — the continuing run’s first objective reproduces the previous run’s last to six digits — but BLMVM’s stored curvature pairs do not survive it, so a chain of jobs is not step-for-step identical to one long run. On a twin at the Houston mesh the restart was not a penalty: a chained $2+2$ iteration run reached a lower objective than a single cold four-iteration run, the restart acting as a rescaling of the quasi-Newton model. That is one twin, not a rule; the point for planning is that chaining did not cost convergence there.

6 Implicit stepping, enabled and required

The explicit drag term is stiff in thin water: linearizing gives a stability limit $\Delta t \lesssim 2h^{4/3}/(gn^2|v|)$, which for sheet flow at centimeter depths is seconds — far below the gravity-wave CFL limit of kilometer-scale meshes. Our Hurricane Harvey test configuration (Houston at 1 km, $\Delta t = 30$ s) confirms this empirically: the explicit source treatment produces NaN within the first objective evaluation, in quantitative agreement with the analysis.

The assembled exact Jacobian makes the remedy nearly free: backward Euler via PETSc’s theta method, with SNES driven by the same Jacobian, carries the Harvey window at $\Delta t = 30$ s, and the

Algorithm 1 Strong-constraint 4D-Var calibration of the Manning field, as implemented with TAO/BLMVM and TSAdjoint. The window map $M_{t_{k-1} \rightarrow t_k}$ is the composition of m steps of the discrete forward stepper (explicit RK, backward Euler, or ARK-IMEX); Algorithm 2 differentiates exactly this stepper, so the gradient is exact to roundoff (and inner-solver tolerance, Section 4) for the computed trajectory.

Require: prior/initial guess n_0 , bounds $[n_{\min}, n_{\max}]$, observations y_k at times $t_k = km \Delta t$ ($k = 1, \dots, K$), gauge operator H , error σ , Tikhonov weight β , gradient tolerance ε_g , iteration budget $I_{\max}$

- 1: $n \leftarrow n_0$
- 2: **repeat**
- 3: set the model’s cellwise drag coefficient from $n \triangleright$ identity for per-cell n ; constant over each region or land-cover class for the reduced parameterizations
- 4: $u \leftarrow u_0$ $\triangleright$ fixed initial state (strong constraint)
- 5: **for** $k = 1, \dots, K$ **do** $\triangleright$ forward sweep; trajectory checkpointed — every step, or a revolve schedule for long windows (Section 6)
- 6: $u \leftarrow M_{t_{k-1} \rightarrow t_k}(u; n)$
- 7: $r_k \leftarrow Hu - y_k$
- 8: **end for**
- 9: $J \leftarrow \frac{1}{2\sigma^2} \sum_{k=1}^K \|r_k\|^2 + \frac{\beta}{2} \|n - n_0\|^2$
- 10: $(\lambda, \mu) \leftarrow \text{ADJOINTSWEEP}(\{r_k\})$ $\triangleright$ Algorithm 2: $\mu = dJ_{\text{mis}}/dn$
- 11: $g \leftarrow \mu + \beta(n - n_0)$
- 12: $n \leftarrow \text{BLMVMSTEP}(n, J, g)$ $\triangleright$ Algorithm 3
- 13: **until** $\|g^P\| \leq \varepsilon_g$ or $I_{\max}$ iterations
- 14: **return** n

Algorithm 2 ADJOINTSWEEP: segmented discrete-adjoint backward sweep (TSAdjointSolve with TSAdjointSetSteps). Φ_s denotes the forward map of step s , $u_s = \Phi_s(u_{s-1}; n)$; its transposed Jacobians are built from the assembled RHS Jacobian $\partial f/\partial u$ (Section 3) and the parameter Jacobian $\partial f/\partial n$ according to the integrator’s discrete-adjoint formula.

Require: checkpointed trajectory $\{u_s\}_{s=0}^{Km}$, residuals $\{r_k\}_{k=1}^K$ at steps $s = km$, gauge operator H , error σ

Ensure: $\lambda = dJ_{\text{mis}}/du_0$, $\mu = dJ_{\text{mis}}/dn$

- 1: $\lambda \leftarrow \sigma^{-2} H^T r_K$ $\triangleright \lambda$ holds $\partial J_{\text{mis}}/\partial u(t_s)$ throughout
- 2: $\mu \leftarrow 0$
- 3: **for** $s = Km, \dots, 1$ **do**
- 4: restore u_{s-1} (and stage values): read the checkpoint, or recompute forward from the nearest one when the checkpoint schedule demands
- 5: $\mu \leftarrow \mu + (\partial \Phi_s/\partial n)^T \lambda$ $\triangleright$ accumulate this step’s Manning dependence
- 6: $\lambda \leftarrow (\partial \Phi_s/\partial u_{s-1})^T \lambda$ $\triangleright$ backward model step (transposed step Jacobian)
- 7: **if** $s - 1 = km$ for some $1 \leq k < K$ **then**
- 8: $\lambda \leftarrow \lambda + \sigma^{-2} H^T r_k$ $\triangleright$ observation jump: u_{km} enters J_{mis} directly
- 9: **end if**
- 10: **end for**
- 11: **return** (λ, μ)

Algorithm 3 BLMVMSTEP: one iteration of TAO’s bound-constrained limited-memory variable-metric method [16]. $P_{[n_{\min}, n_{\max}]}$ is componentwise clipping to the bound box; the curvature pairs $\{(s_i, y_i)\}$ persist across calls. The projected gradient used in the convergence test of Algorithm 1 is $g_i^P = g_i$ for n_i strictly inside the box, $\min(g_i, 0)$ at the lower bound, and $\max(g_i, 0)$ at the upper bound.

Require: iterate n , objective J , gradient g , stored pairs $\{(s_i, y_i)\}$, bounds $[n_{\min}, n_{\max}]$

- 1: $d \leftarrow -B g^P$ $\triangleright B \approx (\nabla^2 J)^{-1}$ by the L-BFGS two-loop recursion over the stored pairs
- 2: find $\alpha > 0$ with a projected line search on $n(\alpha) = P_{[n_{\min}, n_{\max}]}(n + \alpha d)$ satisfying sufficient decrease of J $\triangleright$ each trial α costs one forward-plus-adjoint evaluation (lines 3–11 of Algorithm 1)
- 3: $n^+ \leftarrow n(\alpha)$; evaluate g^+ at n^+
- 4: $s \leftarrow n^+ - n$; $y \leftarrow g^+ - g$; store (s, y) if $s^T y > 0$, discarding the oldest pair beyond the memory limit
- 5: **return** n^+

theta-method adjoint passes the same FD gates (Table 1).

We also implemented the sharper tool: an ARK-IMEX splitting with the Manning drag as the stiff implicit part (**temporal:** `arkimex`). The implicit stage system is block-diagonal — an independent 3×3 Newton per cell, no global solve — while fluxes and bed slope remain explicit under the gravity-wave CFL limit. Its adjoint passes the same gates (2.0×10^{-8} for dJ/du_0 , 2.5×10^{-6} for dJ/dn) and it carries the Harvey window at $\Delta t = 30$ s at 6.7 s per forward-plus-adjoint sweep, with gradients matching backward Euler to three digits. The splitting also runs under RDycore’s libCEED operator backend — the explicit side uses the friction-free source Q-functions prepared in PR #359, and the two backends agree to machine precision (5×10^{-16} relative) after 100 ARK-IMEX steps; a device-resident implicit part is the natural follow-up.

Three implementation facts others will hit:

- PETSc’s ARKIMEX adjoint dereferences *both* the implicit- and explicit-part parameter Jacobians, so a zero explicit-part `TSSetRHSJacobianP` must be registered even when the parameter appears only implicitly.
- The embedded step-size controller must be disabled for finite-difference gradient validation: perturbed solves otherwise take different step sequences, which shows up as an apparent $\sim 10^{-4}$ gradient error that vanishes with fixed steps.
- PETSc’s in-memory checkpointing trajectory cannot tolerate a *segmented* forward solve (one `TSSolve` per observation window). `TSSolve` sets its converged reason before the final trajectory write of every call, which the memory trajectory interprets as the end of the forward run, corrupting the checkpoint schedule at every interior observation time. The forward sweep must be a single `TSSolve` with observations recorded by a monitor; the segmented backward sweep is unaffected.

Newton robustness is set by residual discontinuities, not by conditioning. Both items below come from the land-cover-prior runs rather than the smooth- n configurations behind the numbers reported elsewhere, which never trigger either; both fixes are implemented and gated, the boundary fix as an opt-in condition (open issues 4–5). Where implicit stepping fails on this problem, it fails for a reason that is invisible in the smooth twin experiments: the residual is discontinuous, so Newton has nothing to converge *to*. At the land-cover roughness prior ($\bar{n} = 0.10$) rather than the

twins’ smooth $n \approx 0.03$, backward Euler descends cleanly for a dozen iterations and then freezes, with an $O(10^{-13})$ line-search step changing the residual norm by a factor of eight — the signature of a jump, not of ill-conditioning. Two mechanisms are responsible, both catalogued in Section 6.1: the drag’s wet/dry cutoff, whose amplitude scales as n^2 and therefore only becomes disabling at realistic roughness, and the critical-outflow boundary’s stagnation switch. Regularizing the drag velocity removes the first and restores honest convergence over the early transient. The second we removed by replacing the outlet condition itself: a transmissive (zero-gradient) free outflow — the ghost state copies the interior state, the convention of most finite-volume shallow-water codes — is continuous in the interior state by construction, and its ghost map is the identity, so the analytic Jacobian and adjoint carry it exactly (FD-gated like every other block). The effect is not an improvement but a cure: a rain-forced control that Newton could not carry past step 498 under critical outflow (residual pinned at 1.05×10^{-3} relative, tolerance relief buying steps linearly) completes the same window with *zero* failures, 598 of 600 solves converging in two Newton iterations; a one-hour rain-forced calibration window — forward, adjoint, and quasi-Newton line search — runs clean end to end; and a 72-hour rain-forced forward through the Hurricane Harvey crest (259,200 implicit steps on the 2.93M-cell mesh) completes without a single Newton failure. Before the replacement we absorbed the pin by accepting the iterate at a relative nonlinear tolerance of 10^{-3} ; that this was tolerable rather than merely expedient is worth stating precisely: a land-cover-class twin calibration under that setting recovers all fifteen class values to machine precision (objective reduced from 1.5×10^7 to 6×10^{-7} in twenty quasi-Newton iterations), so the gradient path was unimpaired by the loosened inner solve. The general lesson for adjoint retrofits is that nonsmoothness in an operational flood code is not only a differentiability question — it is a nonlinear-solver question, and the two discontinuities that mattered here were both invisible until the parameter field was realistic.

Checkpointed adjoints for long windows. A naive stored trajectory for a full 36-hour Hurricane Harvey window at the 30 m mesh’s CFL-limited $\Delta t = 0.25$ s (518,400 steps, 2.93M cells) would occupy tens of petabytes. PETSc’s checkpointing trajectory (`-ts_trajectory_type` memory with the revolve optimal-schedule library) replaces storage with recomputation. We validated it across the full hierarchy. On the dam-break twin it reproduces the disk-trajectory calibration and gradient gates bit-for-bit for explicit RK, backward Euler, and ARK-IMEX (the implicit integrators need the same tight inner tolerances as in Section 4, since checkpoint recomputation re-solves the stage systems). On the 30 m mesh, a forward-plus-adjoint sweep with only *three* states checkpointed in memory matches the disk-trajectory gradient to all printed digits — and runs slightly faster, because recomputation is cheaper than trajectory I/O at this scale. For the full window, a few hundred checkpoints (~ 14 GB aggregate) bound the recompute overhead by a factor of about four, making the long-window adjoint memory-feasible; the remaining cost is the forward compute itself, addressed next.

6.1 Nondifferentiability catalog

[Reviewers: do we want this subsection at all? It is the least conventional part of the paper. Five of its six entries are now resolved with gated fixes, which is an argument for keeping it as the record of what retrofitting an adjoint onto an operational flood code actually costs — and the counter-argument is that a resolved catalog is engineering detail that belongs in an issue tracker. Cut, keep here, or move to an appendix.]

Retrofitting an adjoint onto an operational flood code surfaces nonsmooth structure worth documenting. Every entry below is annotated with its status: five are resolved and gated in the

test suite, and one remains open.

- **Time step inside the RHS.** (*Resolved.*) Operator-split friction treatments (here: semi-implicit and XQ2018) make f depend on Δt and on intermediate flux divergences; no consistent $\partial f / \partial u$ exists. The remedy is an explicit f plus a genuinely implicit integrator, not a smarter derivative — which is the architectural choice this section exists to justify, and the reason the implicit-versus-explicit cost comparison above is not the one a forward-only code would make.
- **Entropy-fix branches.** (*Resolved.*) The critical-flow fix is piecewise smooth; branch-faithful one-sided derivatives pass FD checks even on transcritical states (Table 1).
- **Critical-outflow stagnation.** (*Resolved by changing the boundary condition.*) The outflow ghost map $h_g = (q^2/g)^{1/3}$ has an unbounded one-sided derivative at $q = 0$, and the inflow branch switches there: a genuine subgradient point. Unlike the others in this list it is not merely a derivative subtlety — the branch switch is a discontinuity of the *residual* itself, since the $u_\perp < 0$ branch zeroes both states (a wall) while the $u_\perp \geq 0$ branch imposes the critical ghost, so the edge flux jumps by the full wet-onto-dry Roe flux, $O(gh^2/2)$, as a boundary cell crosses stagnation. With an implicit integrator this stalls Newton outright: the line search collapses to $\lambda \sim 10^{-6}$ with the residual pinned at a floor of 10^{-4} – 10^{-5} relative. Explicit stepping never sees it. The cure is to change the boundary condition, not the solver: a transmissive (zero-gradient) free outflow — ghost state equal to the interior state, the standard free outflow of finite-volume shallow-water codes — is continuous by construction, has an identity ghost map in the Jacobian, and removes the stall entirely. Provided as an opt-in condition alongside the original, and gated: a 72-hour rain-forced basin forward completed 259,200 implicit solves with zero nonlinear failures.
- **Drag at the wet/dry cutoff.** (*Resolved by regularizing the drag velocity.*) The drag $gn^2 h^{-7/3} \mathbf{q} \|\mathbf{q}\|$ gated at $h_{\min}$ with a plain $\mathbf{q}/h$ velocity is likewise residual-discontinuous: a cell crossing the cutoff with retained momentum switches an $O(h^{-7/3})$ term on and off. The amplitude scales with n^2 , so it is invisible in smooth- n twin experiments and disabling at land-cover roughness. Evaluating the drag with the ANUGA-regularized velocity $\mathbf{u} = \mathbf{q}h/(h^2 + h_{\text{anuga}}^2)$ removes it (the drag then vanishes as $h^{5/3}$), and reduces identically to the plain form at $h_{\text{anuga}} = 0$. The value of h_{anuga} itself is an empirical choice rather than a physical one (open issue 5).
- **Wet/dry branches.** (*Resolved in the derivative; measured as objective nonsmoothness at basin scale.*) The RHS’s dry cutoffs are honored branch-faithfully in every block; fully wet validation cases keep FD comparisons clean, and dry cells are correctly insensitive to their roughness. At basin scale the branches do something no per-block check sees: over long rain-forced windows the trajectory crosses so many per-cell depth-cutoff surfaces that the *objective* is dense with kinks, and a finite-step central difference measures a smoothed slope several percent from the exact one-branch derivative the adjoint returns. This is measured, not conjectured: the gap is 9×10^{-5} (inner-solve floor) over a 60-step window and grows to 10^{-1} -scale over 3600 steps on identical machinery, while the FD value is stable across a decade of probe step (Section 7.2). The practical reading: FD-validate on short windows, where the gate is clean; on long windows the gate measures the objective’s nonsmoothness, not the gradient’s error.

- **The argmax in the peak-WSE observable.** (*Open — the one entry we have not closed.*) The high-water-mark misfit (Section 7.2) depends on the state through $\max_k Hu(t_k)$: piecewise smooth, with kinks where two sampled times tie for a mark’s peak. The adjoint injects each mark’s residual at its recorded argmax time, which is the exact gradient away from ties (a measure-zero set); but the peak time *moves* across optimizer iterates, so the objective the quasi-Newton method sees is only piecewise smooth and its line search can straddle kinks. Practice is untroubled — the FD gates pass away from ties and the twin calibrations descend monotonically — and the instrumented basin-scale gate adds a measurement: across every FD probe on the real 108-mark configuration, *no* mark’s argmax moved and none flipped wet/dry, so at this geometry the argmax kinks are far sparser than the dynamics’ wet/dry kinks (previous entry). This remains the standard footing of max-type observables in calibration rather than a theorem. *Proposed remedy, not yet implemented:* freeze the argmax assignment across an outer iteration, so each inner subproblem is genuinely smooth in the parameters, and re-evaluate the peak times between outer iterations — a fixed point on peak times, converged when the assignment stops changing. This costs one extra forward per outer iteration and makes the smoothness assumption the line search relies on true by construction rather than true in practice. A smooth surrogate (softmax over the sampled times, tempered toward the true maximum) is the alternative, at the cost of biasing the peak it calibrates against.

7 Hurricane Harvey twin experiments: regularization and the real-network limit

The per-cell twin runs on the Houston 1 km Hurricane Harvey mesh (2,746 cells, 4200 s window at $\Delta t = 30$ s). Truth is a two-zone field split at the domain midline, observed at 343 synthetic gauges over 14 observation times, calibrated with implicit stepping and Tikhonov regularization ($\beta = 10^{-5}$). Every cell is wet and moving in this configuration, so all 2,746 parameters are observable. The configuration is deliberately gauge-sparse — 343 gauges for 2,746 parameters, an 8:1 ratio — and it exhibits the classic *semiconvergence* of underdetermined inverse problems (Figure 1). At 300 BLMVM iterations (~ 22 minutes on one laptop core, ~ 4.4 s per forward-plus-adjoint) the recovered map shows the two-zone truth structure cleanly at 26.3% relative L^2 error. Continuing the same weakly regularized optimization to 2000 iterations *degrades* recovery to 38.0%: the misfit keeps decreasing while the parameters drift along observation-null directions, ultimately pinning cells against both bound constraints (Figure 1, center left). Early stopping was acting as the effective regularizer. The behavior disappears when the problem is made well posed: a properly regularized Houston run ($\beta = 10^{-4}$, 686 gauges) reaches 19.4% at a 1000-iteration budget with no drift (Figure 1, center right) — the sharpest recovery of the three, with the zone boundary resolved at the truth midline. This is precisely the overfitting risk documented for distributed friction estimation [7], here reproduced in a controlled twin setting. It motivates the regularization studies (total variation, land-cover priors, discrepancy-principle stopping) planned as follow-on work.

The real gauge network. The synthetic gauge sets above are a controlled ramp; the network that matters is the real one. Twenty USGS gauges fall inside the Houston mesh, seventeen with Hurricane Harvey stage records (HydroShare DOI 10.4211/hs.c037167e497546a1bc1508dfb32a9cff). Re-running the same twin with the observation operator built at those seventeen gauge cells — 238 observations (17 gauges $\times$ 14 times) against 2,746 parameters — is a sobering experiment. BLMVM drives the misfit down by a factor of 2×10^7 , an essentially perfect fit of the observations,

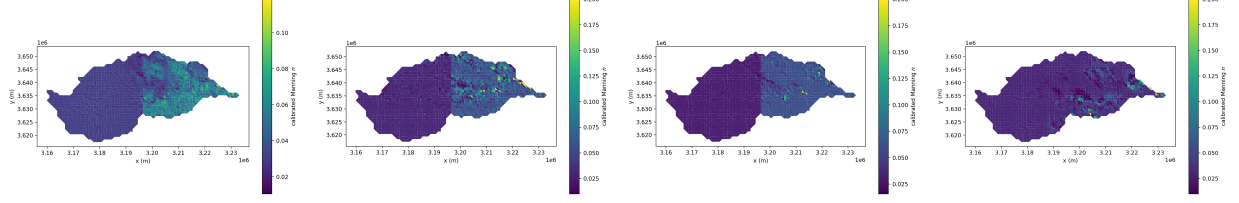


Figure 1: Regularization, semiconvergence, and network density on the Houston twin (2,746 parameters; truth: $n = 0.03$ west, $n = 0.06$ east of the midline; implicit stepping, 14 observation times). Far left: weak regularization ($\beta = 10^{-5}$, 343 gauges) stopped early at 300 iterations — 26.3% error. Center left: the same setup run to 2000 iterations — the misfit is lower but the field drifts along unobserved directions to 38.0% error and saturates the bounds $[0.01, 0.2]$. Center right: proper Tikhonov weight ($\beta = 10^{-4}$, 686 gauges), 1000 iterations — 19.4% error, no drift, sharp zone boundary. Far right: the *real* network — the 17 in-mesh USGS gauges with Harvey stage records, same $\beta = 10^{-4}$, 1000 iterations — the misfit falls 2×10^7 -fold while the field recovers to only 44.9%: no trace of the two-zone truth, near-prior almost everywhere with null-space structure pinned against both bounds.

while the recovered field is 44.9% from the truth and saturates both bounds (Figure 1, far right). This is semiconvergence in its purest form: the real network’s null space is so large that data fit says almost nothing about parameter recovery, and no Tikhonov weight or stopping rule rescues a per-cell field from 238 observations. The practical conclusion for real-data calibration is to shrink the parameter space to match the network — but, as the next subsection measures, shrinking the parameter space is necessary and not sufficient: whether a reduced parameterization is identifiable depends on the observation geometry and window, not on the ratio of observation to parameter counts.

7.1 Identifiability is a property of the observable: gauges versus high-water marks at 30 m

The natural reduced parameterization for this domain is one Manning value per NLCD land-cover class (15 classes at the 30 m Turning mesh, 2.93M cells), with the published NLCD lookup as prior and truth for twins. Counting is deceptively favourable: even the sparse real network supplies an order of magnitude more observations than parameters. Measurement says otherwise. In a class twin on the 30 m mesh at the real 13-gauge geometry (260 observations, a short early-transient window), the objective falls $128\times$ while the recovered class values remain 66% wrong in relative L^2 , worst class 81% — and the failure is structural: classes whose cells are dynamically distant from every gauge (the three forest classes) never move from the starting value, because within the window nothing their roughness does can reach a gauge. The identical configuration observed at 418,076 strided cells recovers all fifteen classes to machine precision, so the failure is the network, not the method. Fifteen parameters, thirteen gauges, and an “overdetermined” observation count: still unidentifiable.

The same question re-asked with the peak observable comes out differently. Observing the same NLCD-truth twin at the 108 QC-passed high-water-mark cells (peak WSE sampled over a one-hour rain-forced window, implicit stepping, free-outflow outlet; Section 7.2) and calibrating fifteen classes from a uniform start: the objective falls $81\times$, the peak-WSE error at the marks drops from 0.103 m to 0.017 m MAE, and the area-dominant classes are identified — the three

Configuration	Signal	Observation noise	Outcome
Verification twin (planar, 40 steps, 1.8 s)	5 μm water height	none 0.1 mm 1 mm	per-region recovery 4×10^{-7} per-cell recovery destroyed ($0.20 \rightarrow 1.8$) per-region destroyed ($4 \times 10^{-7} \rightarrow 0.83$, both at bounds)
30 m basin, 1 h rain-forced, 108 marks	~ 0.1 m peak WSE	none 0.15 m (HWM grade)	area-weighted rel. L^2 0.20; MAE 0.103 $\rightarrow$ 0.017 m misfit falls 9.4 $\times$, MAE 0.23 $\rightarrow$ 0.12 m — but 5 of 15 classes p

Table 2: Parameter-induced observation signal versus observation error. The noise-free recoveries differ by seven orders of magnitude in the signal they descend on; only the basin-scale row is a fair test against survey-grade error (0.1–0.3 m for high-water marks). Noise is a deterministic, decomposition-independent perturbation of the twin observations, so the comparison is reproducible across rank counts.

developed-intensity classes, pasture, and crops recover to 0.7–10.5%, together 83% of the domain’s cells; area-weighted relative L^2 error is 0.20 (unweighted 0.58). The residual errors concentrate in small-area classes (forests, shrub, emergent wetland, $\sim 5\%$ of cells), which now *move* — they are sensitive, unlike in the gauge experiment — but overshoot to the bounds along weakly constrained directions: ordinary semiconvergence, treatable with regularization and longer windows rather than a structural absence of signal. Three caveats keep this honest: the one-hour window samples early-transient peaks, not flood peaks (the real calibration windows are 6–12 hours and should only help); the run was capped at 15 quasi-Newton iterations; and twin marks synthesized from the truth trajectory carry no representation error, so this is an identifiability statement — an upper bound on real-data performance, not a forecast of it.

Noise turns identifiability into a signal-to-noise question. Every recovery above is noise-free, as are the verification twins behind Section 5. Such experiments establish that the gradient is right and that the observable carries the information; they do not establish that a field can be calibrated from real data. What separates the two is the size of the parameter-induced signal relative to observation error, and that ratio moves by orders of magnitude with the configuration (Table 2). On a short verification window the entire Manning-induced observation signal is a few microns of water height, so a noise-free optimizer descends on it indefinitely while a tenth of a millimetre of observation noise destroys the recovery outright — and no Tikhonov weight in a sweep up to $\beta = 1$ rescues it. At basin scale the same quantity is ~ 0.1 m of peak water-surface elevation after one simulated hour, five orders of magnitude larger, which is the regime in which survey-grade high-water-mark error becomes a fair fight rather than a foregone conclusion. The lesson is not that the method is fragile: it is that friction must act long enough to imprint a signal above the noise floor, which makes window length a first-class calibration decision rather than a cost parameter. The basin-scale noisy row shows what marginal signal-to-noise looks like in practice: with ~ 0.1 m of signal against 0.15 m of noise the optimizer still cuts the misfit 9.4 $\times$ — while driving a third of the classes onto their bounds (pasture, 428k cells, to the 0.30 ceiling). The misfit improves because the noise is being fit; the bound-pinning is the visible symptom, and it is the concrete argument for stronger regularization and a longer window in the production run.

7.2 Changing the observable: high-water marks

Calibration against the actual stage records is blocked by geometry, not software: at all 13 rain-driven USGS gauges in the 30 m domain, modelled water-surface elevation sits 1–11 m *above* ob-

served stage, because a 30 m cell stores one mean ground elevation — essentially the bank of an incised bayou — while the gauge hangs in the channel below it; the offset is time-varying as the real channel fills, so no per-gauge datum correction can repair it. We therefore changed the observable to the USGS high-water-mark archive: surveyed *peak* water-surface elevations on structures and debris lines, i.e. on the floodplain surfaces a 30 m cell does represent, and the benchmark currency of the flood-modeling literature (Inunda reports 0.67 m MAE against Harvey high-water marks [8]). Of 2,364 Harvey marks, 324 fall in the Turning domain; applying to the marks the same QC that disqualified the gauges — is the surveyed elevation above the containing cell’s bed? — rejects 62% of them (they are channel surveys), leaving 108 marks of quality fair or better. The misfit compares each mark cell’s peak simulated WSE over the window, $\max_k Hu(t_k)$, against the surveyed value; its adjoint needs no new machinery, because each mark’s residual can be injected at its own argmax time in the segmented backward sweep of Algorithm 2, which is the exact gradient wherever the peak time is locally constant (Section 6.1). The parameter gradient of the peak misfit passes the same central-difference gates as every other block on the smooth verification twin (9×10^{-7} against a 10^{-5} gate). At basin scale the naive check fails — up to 2×10^{-1} on a domain-wide direction — and a set of instrumented measurements pins down why, in a way that validates the gradient rather than the check. First, an instrumented gate that counts argmax movement and wet/dry flips at the marks finds *zero* of either across every probe: the discrepancy lives below the observable. Second, the evaluations are reproducible to nine digits and the FD value is *stable across a decade of probe step* while disagreeing with the adjoint — excluding noise, curvature truncation (ε^2), and sparse kinks (ε^1). Third, and decisively, the gap *accumulates with trajectory length*: on the same basin, rain, marks and machinery, the domain-wide direction errs by 9×10^{-5} (the inner-solve floor) over a 60-step window, 8×10^{-3} over 300 steps, and 10^{-1} -scale over 1800–3600 steps, while single-class directions decay cleanly with probe step. A proportional defect in $\partial f / \partial n$ or the adjoint accumulation would err the same fraction at any window length; trajectory-accumulated branch crossings — the probe shifts which cells cross the depth cutoff at some step — predict exactly this profile. The measured conclusion: the adjoint returns the exact one-branch derivative; over long windows the objective is dense with wet/dry branch surfaces, and a finite-step central difference converges to a locally *smoothed* slope that sits several percent from any one branch. The quasi-Newton iteration consumes one-branch derivatives of a piecewise-smooth objective — standard practice for max-type and switching systems — and its observed descent (Section 7.1) is consistent with that footing (Section 6.1). The remaining scientific choice — confirmed or corrected by the domain team — is the calibration window, which for marks that record the event maximum must cover the flood peak.

The uncalibrated baseline against the real survey. Because the marks record the event maximum, the honest baseline evaluation must simulate through the crest: a 72-hour rain-forced forward from the available initial state (2017-08-26 18:00, reaching past the Buffalo Bayou crest), NLCD-prior Manning, free outflow, backward Euler at $\Delta t = 1$ s on the 2.93M-cell mesh — 259,200 implicit steps, *zero* Newton failures, two hours on one GPU node. The model wets all 108 marks, and the uncalibrated peak-WSE MAE against the surveyed values is 3.41 m (confirmed unchanged, mark for mark, at fully converged inner tolerances). The per-mark decomposition splits that number into two distinct populations. Seventy-one marks crest inside the window; against them the model reads 1.51 m MAE with a +1.20 m median-high bias — error a Manning field can plausibly address. The other 37 never crest at all: their cells fill monotonically through all 72 hours — *every one* peaks at the final hour with exactly zero recession, still rising at 0.02–0.06 m/hr while the other population is already draining — accumulating a mean 10.7 m of standing water, +7.1 m above every surveyed

value. Those cells sit in the downstream Buffalo Bayou reach, and their late-window inflow arrives laterally after the rain has ended: the model cannot drain that reach, a conveyance/outlet defect that no roughness value can absorb and that would corrupt any calibration asked to fit it. The calibration target is therefore the 71 admissible marks and their 1.51 m, the drainage defect is reported as a finding in its own right, and the remaining bundle (initial-condition, rainfall, mesh-elevation, and datum error) is what the calibration must be prevented from absorbing along with the roughness signal. The literature’s calibrated benchmark on this event is 0.67 m MAE [8].

8 Conclusions and outlook

An exact, verified Jacobian turns out to be the single artifact from which everything else follows cheaply: adjoint state and parameter sensitivities, gradient-based calibration, and implicit stepping. The capability is production-ready in the sense that matters for an operational code — behind a flag, FD-gated in CI, and with the verification baseline (FD coloring) kept alive in the configuration system.

Next steps, in order: the Houston per-cell demonstration with real USGS Harvey gauges (datum QC pending); regularization studies (total variation, land-cover priors) for the ill-posedness that our observability results quantify; and coupling the gradient path to PyTorch through the existing PETSc–PyTorch bridge for learned-physics and hybrid data-assimilation experiments.

A closing note on cost. Checkpointing makes the long-window adjoint memory-feasible, but at the 30 m mesh’s CFL-limited $\Delta t = 0.25$ s a full 36-hour Hurricane Harvey gradient still costs several forward sweeps over half a million steps. The affordable path stacks reductions that are all now in place. First, fully implicit stepping at CFL-free time steps: backward Euler with the assembled Jacobian carries the 30 m, 2.93M-cell Harvey-region mesh at $\Delta t = 1$ s — four times the explicit limit, with Newton converging in about four iterations per step behind a block-Jacobi-preconditioned Krylov solve; larger implicit steps are under investigation. Second, GPU execution of the entire gradient pipeline: the right-hand side, the assembled Jacobian and its transpose solves, the preconditioner setup, the parameter Jacobian, the observation operator, and the revolve-checkpointed trajectory are all device-resident. On four A100 GPUs, a forward-plus-adjoint gradient of a one-simulated-hour window on that mesh (3,600 implicit steps, 400 in-memory checkpoints) takes about nine minutes, and one TAO iteration of a dense-gauge twin calibration (418,076 gauges, 39-step window) takes 1.7 s — roughly $19\times$ one 64-core CPU node of the same system. Device and host runs produce bitwise-identical trajectories under a common compiler and gradients identical to every printed digit, so the finite-difference gates of Table 1 carry over unchanged. Third, land-cover-class parameterization (tens of TAO evaluations instead of hundreds) and calibration windows placed on the storm peak rather than the full event.

Red/tiger-team review: the weakest parts of this draft

[Temporary section, removed before submission. This is our own adversarial read of the draft — the places a hostile reviewer (or a careful friend) will push, ranked by how much damage they could do. Most map directly onto the numbered issues in the next section; cross-references are given as *Qn*.]

1. **There is no real-data result in the paper.** The abstract’s differentiator — production code made differentiable, versus reduced solvers rewritten in ML frameworks — invites the comparison the paper cannot yet win: Inunda reports 0.67 m MAE against *surveyed* Harvey

high-water marks; every calibration here is a twin. The real-gauge experiment we do report (17-gauge network, misfit down 2×10^7 -fold, field 45% wrong) is honest and instructive, but it is a negative result about the network, not a demonstration of skill. Until a real-observation number exists, the introduction’s framing overdraws. *Mitigation in progress:* the USGS high-water-mark archive is acquired and QC’d (2,364 marks; 324 in-domain; 108 usable after the below-bed and quality filters), a peak-WSE misfit with an exact argmax-time adjoint is implemented and FD-gated, the identifiability twin at the real 108-mark geometry appears in Section 7.1, and the *uncalibrated* real-data baseline is now measured and decomposed (Section 7.2: 3.41 m MAE over a crest-covering 72-hour forward; 1.51 m over the 71 marks in reaches the model can drain, with the other 37 isolated as a conveyance defect). The checkpointed mid-event initial state the calibration window needs is implemented and validated bit-exactly. Still missing: only the calibrated number itself, which waits on the window-placement decision. Until then the paper claims the capability, the baseline and its decomposition, not the closed loop. (Q1, Q2)

2. **The semiconvergence cure is asserted by a counting argument our own evidence undercuts.** Section 7 proposes the land-cover-class parameterization because 238 observations “overdetermine” ~ 13 classes $18\times$. But observation *count* is not information content: in an internal class-parameterization twin at the real 13-gauge geometry (short window), the objective fell $128\times$ while the class values finished 66% wrong — three classes never moved because no water reachable from their cells crosses a gauge in the window. The honest statement is that identifiability depends on the observation *geometry and window*, and must be measured, not counted. *Addressed:* the counting argument is gone; Section 7.1 now reports the measured study — 13 gauges fail (classes distant from every gauge never move), 108 high-water marks identify the classes covering 83% of the domain — with its caveats (short window, iteration cap, no representation error) stated. Remaining exposure: the caveats themselves; a reviewer may ask for the 6–12-hour-window version, which is exactly the planned production run. (Q2)
3. **The verified gradients are not the production gradients.** Every FD gate in Table 1 is passed at tight inner tolerances; the planned production configuration runs the inner Krylov solver at $\text{rtol } 10^{-2}$, where the adjoint-vs-FD mismatch is inner-solve-limited at 6×10^{-4} – 4×10^{-3} . In every test this does not affect BLMVM convergence. *Partially addressed:* Section 4 now states the production tolerance, the measured gradient error, and the spot-check policy. Still open: a principled footing (inexact-gradient quasi-Newton has known sufficient conditions) rather than an empirical one. (Q3)
4. **“Implicit stepping is required” is conditional on a choice we made.** The production code handles drag stiffness with operator-split friction treatments; they are unusable for the adjoint because they bury Δt inside the RHS, and *that* is why the adjoint path needs an explicit source plus an implicit integrator. The section title claims more than this. Also missing is the honest wall-clock comparison at 30 m: backward Euler at $\Delta t = 1$ s costs ~ 4 – 5 Newton iterations per step behind a Krylov solve versus explicit RK at the CFL-limited 0.25 s — the net factor is modest and should be printed, not implied. Larger implicit steps ($\Delta t \geq 5$ s), which would make the case decisive, are currently blocked by a linear-solver failure the draft does not mention. *Partially addressed:* the comparison was run, and the premise was wrong — explicit stepping of the *differentiable* right-hand side diverges at 0.25, 0.125 and 0.05 s, because the Δt -free drag is stiff on thin films ($\Delta t \lesssim 2.4$ ms) well below the CFL limit. The modest factor holds only against the operator-split discretization, which is not the one

being differentiated; both readings are now stated in Section 6. The $\Delta t \geq 5$ s linear-solver failure is now stated there too. Still open: making $\Delta t \geq 5$ s work. (Q6)

5. **We changed the physics to fix the solver, and the changes await domain sign-off.** The regularized drag velocity ($h_{\text{anuga}} = 0.001$ m, chosen as “smallest that cures the stall”; 0.01 m observed unstable and the mechanism only conjectured) and the transmissive outlet (replacing critical outflow; alters outlet physics; weakly reflective for subcritical flow) are both defensible and both flagged opt-in, but a reviewer can fairly ask whether the calibrated n compensates for boundary and regularization choices rather than physics. Sensitivity of the calibrated field to h_{anuga} and to the outlet choice should be demonstrated, not asserted. (Q4, Q5)
6. **The Hurricane Harvey section is a 2,746-cell twin.** As originally drafted, all 30-m-mesh, 2.93M-cell content was cost measurement — no calibration at 30 m was shown, a gap a reviewer comparing against the 471M-cell forward-validation claim in the introduction would notice. *Addressed:* the section is retitled, and Section 7.1 and Table 2 now contain three 30 m, 2.93M-cell class calibrations run end to end on one GPU node (the 13-gauge twin, the 108-high-water-mark twin, and the noisy HWM twin). What remains true: no *real-data* calibration at any resolution (item 1), and no *per-cell* calibration at 30 m — only the class parameterization, which the identifiability evidence says is the right one anyway.
7. **All twins are noise-free.** σ enters only as a scale factor; no experiment adds observation noise, so the semiconvergence story (early stopping, Tikhonov weights) is undisturbed by the noise level that in practice sets where semiconvergence turns over. *Partially addressed:* a deterministic noise option now exists and Table 2 sets the parameter-induced signal against the observation error for both regimes — reframing the noise-free recoveries as machinery verification and the noise question as a signal-to-noise (window length) question. The noisy basin-scale HWM twin is now measured (misfit falls 9.4 $\times$ while five classes pin to bounds — Table 2 and Section 7.1). Still open: noise-consistent regularization (discrepancy-principle stopping). (Q7)
8. **The peak-WSE observable, if adopted, is nonsmooth in a new way.** The argmax-time adjoint is exact wherever the peak time is locally constant, and ties have measure zero — but the peak time *moves* across optimizer iterates, so J is only piecewise smooth and BLMVM’s line search can straddle kinks. Practice (and our FD gates away from ties) supports it; theory does not. *Addressed:* the nondifferentiability catalog now carries this entry explicitly; the theoretical footing remains open by construction. (Q2, Q3)
9. **Reproducibility currently rests on an unmerged PETSc fork.** The device-resident assembled-Jacobian path (blocked-COO BAIJ assembly on GPU) exists only on our PETSc branch. Until it is upstreamed or the paper’s results are reproducible with stock PETSc (AIJ path, at some cost), the artifact story is weaker than the “behind a flag, FD-gated in CI” framing suggests. (Q8)
10. **The novelty claim needed — and got — a literature sweep.** The original “first E3SM component model with built-in discrete-adjoint parameter estimation” was wrong as stated: MALI routinely inverts for basal friction with an automatic-differentiation adjoint of its Albany velocity solver [14, 13]. The claim is now narrowed to the first *transient* discrete adjoint (differentiated through the time integrator) in an E3SM component, with MALI cited in the abstract and introduction. Remaining check before submission: ocean/MPAS

adjoint efforts, and whether any E3SM land-model calibration uses a true adjoint rather than ensemble or surrogate gradients.

Open issues and current positions, in order of importance

[Temporary section, removed before submission. These are the open issues, in the order they matter to the paper, tagged with the team each belongs to: science (Gautam, Donghui, and colleagues), numerics (Matt, Jed), engineering (Jeff). Each states what we found, what we did, and the position we are proceeding from — we are not blocked on any of them. Nomenclature: “WSE” is water-surface elevation (bed elevation plus water depth), which is what a stage gauge reports once put on the NAVD88 datum; “cell bed” is the single mean ground elevation our 30 m mesh stores for a cell; “HWM” is a USGS-surveyed high-water mark, the peak WSE reached at a point during the event.]

1. **[Science] The real-data calibration stands on high-water marks, not stage.** The stream gauges sit in channels our mesh cannot see: at all 13 rain-driven USGS gauges in the Turning domain, modelled WSE sits *1–11 m above* observed stage, only 1 of 13 gauges has observed water above its own cell’s bed on the rising limb, and the offset is time-varying (9.7 m $\rightarrow$ 1.7 m over twelve hours at Whiteoak/Main St, where our cell stays essentially dry) — so no fixed per-gauge datum correction can work. *What we did:* acquired the USGS Harvey high-water-mark archive (2,364 marks; 324 in the Turning domain), applied the same below-bed QC that disqualified the gauges (62% of in-domain marks fail it — they are channel surveys), keeping 108 marks at quality $\leq$ fair, and implemented a peak-WSE misfit (exact argmax-time adjoint, FD-gated). HWMs record the peak, sit on floodplains and structures the 30 m cell can represent, need no hydrograph timing, and are the benchmark currency of the field (Inunda: 0.67 m MAE). *Position:* peak WSE at the marks is the primary observable. Gauge stage stays available as a secondary misfit restricted to times when model and observation are both above the cell bed, and flood-extent (wet/dry) agreement is a candidate third; neither is needed for the headline number.
2. **[Science + numerics] Which parameterization the data actually supports.** The evidence: 238 gauge observations fit essentially perfectly while a per-cell field is 45% wrong; a class-parameterized twin at the real 13-gauge geometry (short window) drove the objective down $128\times$ with class values still 66% wrong; the same experiment at the 108 HWM locations identifies the classes covering 83% of the domain (Section 7.1). The draft’s counting argument (“overdetermined $18\times$ ”) is indefensible and is gone. *Position:* the identifiability study — which classes are observable from which networks and windows — is a paper contribution in its own right, and NLCD-class with Tikhonov to the NLCD prior is the production parameterization.
3. **[Numerics] Inexact gradients at production solver tolerances.** At the campaign setting (`gmres`, right preconditioning, `ksp_rtol` 10^{-2} , `pbjacobi`) the adjoint gradient differs from FD by 6×10^{-4} – 4×10^{-3} relative — inner-solve-limited, not a bug; the same configuration at `rtol` 10^{-3} passes the 10^{-5} gates. BLMVM converges indistinguishably in every test (same J to 5–6 digits). *Position:* publish the production tolerance alongside the measured gradient error and the spot-check policy. Principled inexactness control — forcing terms tied to the line-search decrease, and possibly a tighter tolerance on the adjoint solves than the forward ones, since the gradient inherits both — is the obvious refinement and is not done.

4. **[Science] Outflow boundary at zero velocity.** The critical-outflow condition switches abruptly at zero normal velocity: below it the boundary acts as a wall, above it it imposes critical outflow, so the flux jumps by roughly $gh^2/2$ as a boundary cell crosses stagnation, which pins the implicit solver (isolated by controls: a reflecting outlet completes the identical run; $5\times$ stronger friction regularization changes nothing; tolerance relief is a treadmill because the pin level follows the local flow state). We implemented the least clever standard treatment: a *transmissive* (zero-gradient) free outflow — ghost state equal to the interior state, as in ANUGA and Clawpack — continuous by construction, differentiated exactly through host and device Jacobians. It cures the stall: the failing control completes with zero failures and two-iteration Newton solves, and a 72-hour rain-forced forward completes 259,200 implicit solves with zero failures. It is opt-in (`type: free-outflow`); the original remains the default. Caveats: exactly non-reflecting only for supercritical outflow; does not prevent inflow if the interior velocity points inward. *Position:* transmissive outflow for the production runs, with a Froude-limited or blended form as the fallback if those caveats bite on Buffalo Bayou.
5. **[Science] Bed-friction regularization.** The friction term $\tau_b = gn^2h^{-7/3}\mathbf{q}\|\mathbf{q}\|$ switched off below the depth cutoff makes the equations jump as a cell crosses it; the jump grows like n^2 , invisible in the twins but disabling at land-cover roughness ($\bar{n} = 0.10$). We now evaluate the friction with the ANUGA-regularized velocity, so drag fades smoothly to zero with depth. Off by default; runs use $h_{\text{anuga}} = 0.001$ m, the smallest value that cures the stall; 0.003 m is indistinguishable; 0.01 m is unstable (mechanism conjectured: weaker damping lets an iterate reach $h < 0$). *Position:* keep the ANUGA form at $h_{\text{anuga}} = 0.001$ m. The value is an empirical choice, not a physical one, and the sensitivity above is the whole of our evidence for it.
6. **[Numerics] Implicit solver headroom: bigger steps and better preconditioning.** Backward Euler with the assembled Jacobian carries the 2.93M-cell mesh at $\Delta t = 1$ s at ~ 4 –5 Newton iterations per step with `pbjacobi`-preconditioned GMRES; at $\Delta t = 5$ s the very first Newton *linear* solve diverges (300 GMRES iterations, `pbjacobi`) — a preconditioner failure before nonlinearity even enters. A wall-clock ratio against *explicit stepping of the same right-hand side* is not the modest number one might expect: with the Δt -free drag the adjoint requires, forward Euler is limited by the friction rate, $\Delta t < 1/t_b$ with $t_b = gn^2h^{-4/3}|v|$, and not by the CFL. On the spun-up 30 m state (99.9% wet, median depth 6 mm, t_b up to 420 s^{-1}) that is $\Delta t \lesssim 2.4$ ms against a CFL-limited 0.25 s — a factor ~ 400 in step count, in the implicit path’s favour. Confirmed end to end on the rain-forced window: forward Euler returns a non-finite state at $\Delta t = 0.25, 0.125$ and 0.05 s, on configurations differing from the backward-Euler run in the integrator and step size alone, while backward Euler at $\Delta t = 1$ s completes cleanly. The comparison that *would* be modest — explicit stepping at 0.25 s using the operator-split friction — is a comparison against a different, non-differentiable discretization, and we note it as such rather than quoting it as this method’s cost. *Open:* what unlocks $\Delta t \geq 5$ s. The candidates are a stronger preconditioner (AMG on a reduced system, block ILU, field-split), pseudo-transient continuation or another globalization once the linear solve holds, or simply accepting more Krylov iterations; whether to keep reassembling the Jacobian every step is open alongside it.
7. **[Science] Production-run design: window, observation error, gauge trust.** Four choices remain. (a) Which phase of Harvey to calibrate on: the observability study favours the rising limb, while the peak-agreement result and the HWM observable favour the peak; the per-mark crest times now being measured decide it. (b) The observation error σ per observable, which sets the data-versus-prior weight directly — USGS quality codes give 0.1–

0.3 m for HWMs, and we are using 0.15 m. (c) The exclusion of eight reservoir-influenced sites (Addicks/Barker and Buffalo Bayou downstream), on the grounds that gate releases are unmodelled — now backed by a measurement on the mark side: the 37 downstream marks that never crest sit in cells the model demonstrably cannot drain (monotone filling through all 72 hours, zero recession, +7.1 m above every surveyed value; Section 7.2), so scoring or calibrating against them would fold a conveyance defect into the roughness field. (d) The bounds $n \in [0.01, 0.2]$ and the use of the NLCD field as the prior throughout, replacing the two-zone twins’ constant prior.

8. **[Numerics + engineering] Upstreaming the PETSc-side pieces.** The device-resident assembled-Jacobian path relies on our PETSc branch: blocked-COO assembly for BAIJ matrices on GPU (`MatCOOUseBlockIndices`; `baijkokkos/baijcuda` classes) plus small TSAdjoint fixes. We think it is generally useful — one index pair per block is the natural COO for FV systems — and intend to upstream it as is. *Open:* whether the API needs changes before it hardens, and whether the paper’s artifact claim should be scoped to the stock-PETSc AIJ path until it lands.
9. **[Engineering] Public API and CI.** The adjoint driver reaches through `private/rdycoreimpl.h` (the LETKF-driver pattern). *Open:* which calls deserve promotion to `rdycore.h` (`RDySetObservations`, `RDyAdjointSolve`, `RDyGetSensitivities` are the candidates), and whether Fortran bindings are needed at that point. The new tests need a TSAdjoint-capable PETSc and run in seconds; the CEED-only lanes correctly skip `numerics.jacobian` configs.
10. **[All] Venue and authorship.** Target journal (WRR / JAMES / GMD) and the co-author list remain to be settled.

References

- [1] Yan Ding, Yafei Jia, and Sam S. Y. Wang. Identification of Manning’s roughness coefficients in shallow water flows. *Journal of Hydraulic Engineering*, 130(6):501–510, 2004.
- [2] Xianqiang Chen et al. Adjoint model of shallow water equations based on the finite volume method and unstructured meshes. *Abstract and Applied Analysis*, 2014:407468, 2014. Twin-experiment inversion of Manning roughness; velocity vs. stage observation sensitivity.
- [3] Asier Lacasta, Mario Morales-Hernández, Pilar García-Navarro, et al. A GPU accelerated adjoint-based optimizer for inverse modeling of the two-dimensional shallow water equations. *Computers & Fluids*, 2016.
- [4] Jérôme Monnier et al. DassFlow: Data assimilation for free-surface flows (DassHydro platform). <https://dasshydro.github.io>, 2025. 2D finite-volume shallow water with variational DA; Manning, bathymetry, and discharge inference incl. SWOT/AirSWOT and video-derived surface velocity.
- [5] Pierre Brisset, Jérôme Monnier, Pierre-André Garambois, and Hélène Roux. River discharge estimation from synthetic SWOT-type observations using variational data assimilation and the full Saint-Venant hydraulic model. *Journal of Hydrology*, 558:176–195, 2018.
- [6] Tuomas Kärrnä et al. Efficient optimization of a regional water elevation model with an automatically generated adjoint. *Journal of Advances in Modeling Earth Systems*, 15, 2023. Spatially varying Manning friction calibrated for the Baltic Sea with Thetis/Firedrake/pyadjoint.

- [7] Simon C. Warder, Kevin J. Horsburgh, and Matthew D. Piggott. Adjoint-based sensitivity analysis for a numerical storm surge model. *Ocean Modelling*, 160:101766, 2021.
- [8] Zhi Li. Inunda: A GPU-native, agent-enabled, differentiable solver for high-resolution flood inundation modeling. arXiv:2607.09614, 2026. PyTorch local-inertial solver; roughness/bed/infiltration by gradient descent; Hurricane Harvey hindcast, Harris County TX.
- [9] Xiaofeng Liu et al. Scientific machine learning of flow resistance using universal shallow water equations with differentiable programming. *Water Resources Research*, 61, 2025. Hydrograd (Julia/SciML): AD-based Manning inversion and neural flow-resistance terms.
- [10] Elsa Cardoso-Bihlo and Alex Bihlo. A solver-in-the-loop framework for end-to-end differentiable coastal hydrodynamics. arXiv:2604.07129, 2026. AegirJAX: differentiable non-hydrostatic shallow-water solver in JAX.
- [11] Shengyu Kang, Jiabo Yin, and Dai Yamazaki. CaMa-Flood-GPU: A GPU-based hydrodynamic model implementation for scalable global simulations. *Geoscientific Model Development*, 19:5623–5640, 2026.
- [12] Tadd Bindas et al. Improving river routing using a differentiable Muskingum-Cunge model and physics-informed machine learning. *Water Resources Research*, 60, 2024. Differentiable routing inferring Manning roughness and channel geometry with neural networks.
- [13] Matthew J. Hoffman, Mauro Perego, Stephen F. Price, William H. Lipscomb, Tong Zhang, Douglas Jacobsen, Irina Tezaur, Andrew G. Salinger, Raymond Tuminaro, and Luca Bertagna. MPAS-Albany Land Ice (MALI): a variable-resolution ice sheet model for Earth system modeling using Voronoi grids. *Geoscientific Model Development*, 11(9):3747–3780, 2018.
- [14] Mauro Perego, Stephen Price, and Georg Stadler. Optimal initial conditions for coupling ice sheet models to Earth system models. *Journal of Geophysical Research: Earth Surface*, 119(9):1894–1917, 2014.
- [15] Hong Zhang, Emil M. Constantinescu, and Barry F. Smith. PETSc TSAdjoint: A discrete adjoint ODE solver for first-order and second-order sensitivity analysis. *SIAM Journal on Scientific Computing*, 44(1):C1–C24, 2022.
- [16] Steven J. Benson and Jorge J. Moré. A limited memory variable metric method in subspaces and bound constrained optimization problems. Technical Report ANL/MCS-P909-0901, Argonne National Laboratory, Mathematics and Computer Science Division, 2001.